

# AMARNATH C R

Lead Platform Engineer | Kubernetes & Cloud-Native Infrastructure | DevOps & GitOps

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## PROFESSIONAL SUMMARY

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Senior Platform Engineer and technology leader with 13+ years of progressive engineering experience, currently leading the design and delivery of Kubernetes-based developer platforms at Deloitte. Proven track record of bridging deep application-engineering expertise (Java, Spring Boot, Python) with modern cloud-native infrastructure (Kubernetes, ArgoCD, Terraform, Helm) to build internal developer platforms that accelerate engineering velocity. Certified Kubernetes Application Developer (CKAD) with hands-on experience across AWS, GCP, and Azure. Adept at technical leadership, architectural decision-making, and mentoring engineers across distributed teams.

## CORE COMPETENCIES

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**Container Orchestration & Platform:** Kubernetes (CKAD), Helm, Kustomize, ArgoCD, GitOps, Istio (Service Mesh), KEDA (Event-driven Autoscaling), CastAI (Cost Optimization), Container Networking, Operator Patterns

**Infrastructure as Code & Automation:** Terraform, GitHub Actions, GitLab CI/CD, Jenkins, Circle CI, Docker, npm

**Cloud Platforms:** AWS (EC2, RDS, S3, Lambda, CloudWatch), Google Cloud Platform, Microsoft Azure, SAP BTP (Kyma Runtime)

**Observability & Monitoring:** Prometheus, Grafana, Loki, Promtail, Tempo, Fluent Bit, OpenTelemetry

**Backend & Application Engineering:** Java, Spring Boot, Spring Cloud (Eureka, Zuul, Config Server, OAuth2), Python (FastAPI), Groovy on Grails, Microservices, REST APIs, Junit

**Data & Big Data:** Apache Spark, Apache Airflow, Kubeflow, SQL (Oracle, MySQL)

**Frontend & Mobile:** ReactJS, JavaScript, TypeScript, HTML/CSS, Bootstrap, Material-UI, React Native, Android

**Practices & Methodology:** Microservices Architecture, Design Patterns, OOP, SDLC, Agile, API Design, Code Review, Technical Mentoring

**Tooling:** Git, GitHub, GitLab, JIRA, Confluence, SonarQube, Swagger/OpenAPI

## CERTIFICATIONS

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- **Certified Kubernetes Application Developer (CKAD)** — Cloud Native Computing Foundation (CNCF)

## PROFESSIONAL EXPERIENCE

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### Senior Consultant 2 — Lead Platform Engineer

June 2021 – Present

**Deloitte Consulting India Pvt. Ltd.** | Bengaluru, India

*Lead engineer for Deloitte Engineering Platforms — building internal Kubernetes-based developer infrastructure consumed by engineering teams across the firm. Drove the transition from a senior application engineer into a platform-engineering leadership role, owning architecture, delivery, and team mentoring across two flagship platforms.*

### Kubernetes Management System (KMS) — Lead Developer

- Architected and led the development of the firm's internal Kubernetes management platform, providing self-service cluster provisioning, workload onboarding, and lifecycle management for engineering teams across the organization.
- Designed and implemented GitOps-based delivery pipelines using ArgoCD, Helm, and Kustomize, enabling declarative, auditable, and repeatable deployments across multiple environments and clusters.
- Engineered Infrastructure-as-Code modules in Terraform to standardize provisioning of cloud resources and Kubernetes clusters, reducing onboarding friction and eliminating configuration drift across environments.
- Defined platform APIs, resource abstractions, and operator-driven workflows that allow application teams to consume infrastructure without deep Kubernetes expertise, improving developer experience and adoption.

- Extended the platform onto SAP BTP Kyma Runtime, onboarding workloads to Kyma-managed Kubernetes clusters and integrating Kyma-native eventing, serverless functions, and service bindings into the platform's delivery model.
- Introduced CastAI for Kubernetes cost optimization across managed clusters — enabling automated node right-sizing, bin-packing, and spot-instance adoption to materially reduce cloud spend while preserving workload reliability.
- Led technical grooming, design reviews, and architectural decisions; mentored a team of platform engineers, conducted code reviews, and established engineering standards and best practices.
- Authored comprehensive API documentation (OpenAPI/Swagger) and platform onboarding guides; partnered with consumer teams to drive adoption and resolve integration issues.

#### **CortexAI (Big Data & Analytics Platform) — Senior Developer**

- Contributed to the design and delivery of CortexAI, a big data and analytics platform leveraging Apache Spark, Apache Airflow, and KubeFlow for orchestrated data and ML pipelines on Kubernetes.
- Built and maintained backend services and APIs using Python (FastAPI) and Java/Spring Boot, integrating with cloud-native storage and compute layers across AWS and GCP.
- Implemented CI/CD pipelines using GitHub Actions and GitLab CI for containerized workloads, including automated testing, image scanning, and progressive delivery.
- Provided technical mentoring on cloud-native patterns, container best practices, and microservices design to feature teams; supported integration and automation testing efforts.

#### **Senior Associate — Spring Boot / ReactJS Developer (Fullstack)**

*March 2020 – May 2021*

**Cognizant Technology Solutions (Client: Verizon Wireless) | Coimbatore, India**

- Developed retail-store applications for Verizon Wireless, serving a national footprint of Verizon and partner retail outlets across the United States.
- Built microservices using Spring Boot and the Spring Cloud ecosystem (Eureka, Zuul API Gateway, Config Server, Spring Security OAuth2), integrated with ATG and Oracle backends.
- Containerized services with Docker and contributed to deployment automation; authored Swagger/OpenAPI documentation for service contracts.
- Owned design and development across the stack including ReactJS frontend; supported integration and automation testing, and mentored junior team members.

#### **Senior Technical Lead — Java/J2EE Developer**

*June 2013 – February 2020*

**Amphisoft Technologies (Product: E-Box) | Coimbatore, India**

*Progressed through three engineering roles — from Software Engineer to Senior Technical Lead — building E-Box, an online auto-evaluation platform widely used by enterprise customers (CTS, Payoda) and leading universities (Anna University) for assessments, training, and recruitment.*

- Architected and built monolithic and service-oriented platforms using Java/J2EE, Spring Boot, Groovy on Grails, and ReactJS, supporting auto-evaluation across multiple programming languages (C, C++, Java, OpenCV, Objective-C, Shell, ASP.NET).
- Designed and deployed cloud-native services on AWS (EC2, RDS, S3, Lambda, CloudWatch), including production-cluster scaling and operational monitoring.
- Led the migration from monolith to microservices architecture using the Spring Cloud ecosystem (Spring Boot, Config Server, OAuth2, Zuul Gateway, Eureka).
- Built native Android applications and hybrid mobile experiences using React Native, extending platform reach to mobile users.
- Owned product architecture, technical design, code reviews, and mentoring across the engineering team in the final years of tenure.
- Conducted POCs in image classification using AWS SageMaker and Google Vision (mark-sheet keyword extraction), and explored reactive programming via Spring Boot 2 Reactive Stack.

#### **ADDITIONAL**

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- **Awards:** Winner, State-level Programming Contest (team event)
- **Interests:** Cricket, Classical Music, Continuous Exploration of Cloud-Native and ML/AI Technologies